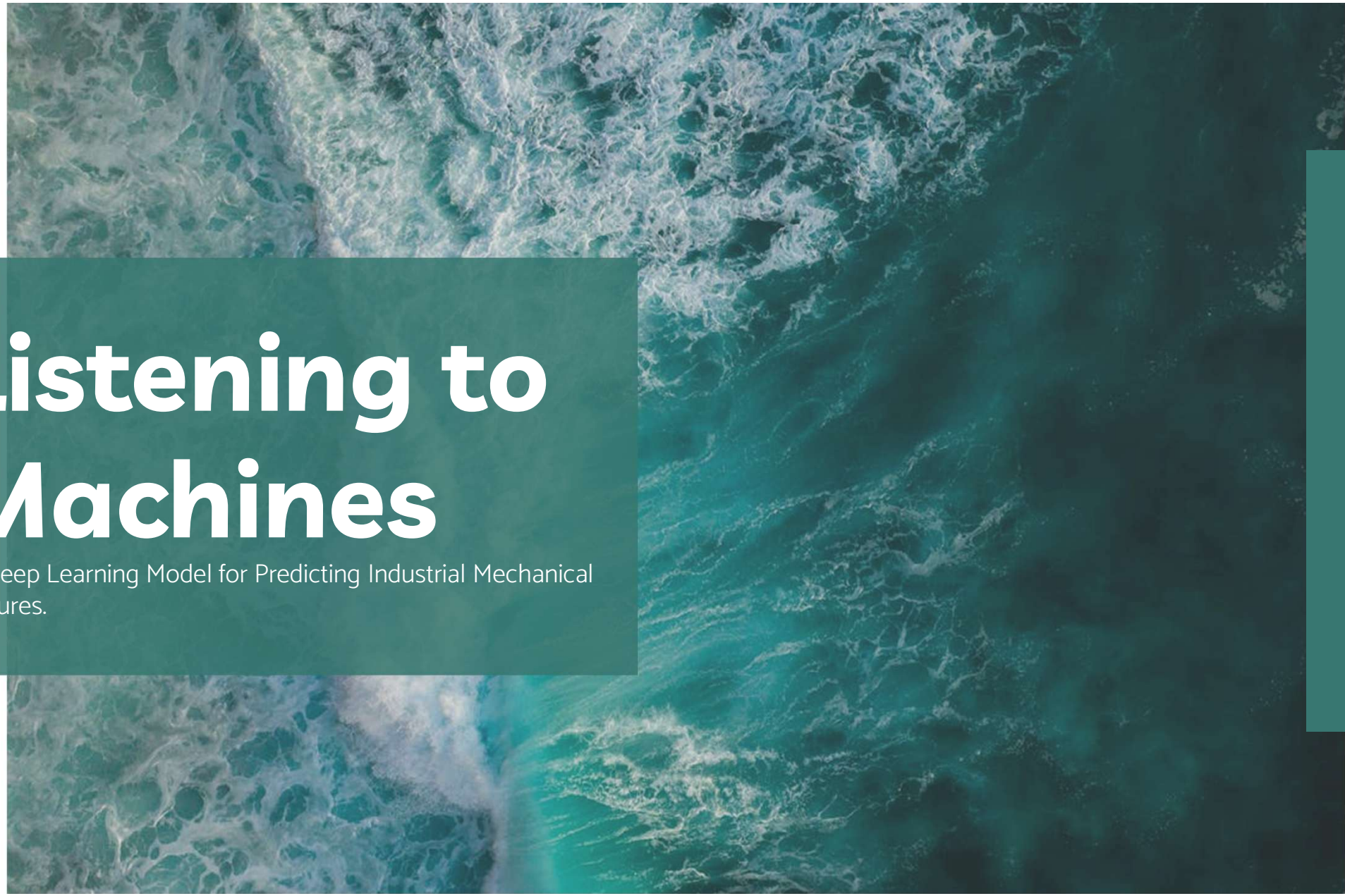

Listening to Machines

A Deep Learning Model for Predicting Industrial Mechanical Failures.

ABOUT THE PROJECT

This Deep Learning project uses a custom CNN to identify mechanical anomalies through sound to prevent costly downtime.

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The Cost of waiting for things to break, and exploratory data analysis.

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06 **Is It Over Yet?**
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PROJECT GOALS

GOAL 1

Build a model that functions in the real world, not just in a quiet laboratory.

GOAL 2

Develop a method to visual prove the model's logic.

The Cost of Waiting for Things to Break

1) **Reactive:** Wait for it break, then fix it. Catastrophic downtime, lost production, secondary damage.
2) **Scheduled:** Service every machine on a fixed calendar. Safer, but wasteful. Sometimes we replace parts that still have months left of life.
3) **Predictive:** Monitor constantly. Intervene only when a failure is imminent. Cheapest in the long run.

Major Concerns

Whisper in a Crowded Room

Audio of the Pumps are mixed with background noise at -6 dB SNR.

The background noise is 4x louder than the pump

The Accuracy Paradox

A “lazy” model that always predicts normal, will score 89% accuracy, despite catching zero failures.

Black Swan

- Normal Recordings: 3,749
- Abnormall Recoding’s: 456

~11%

The Asymmetric Cost

False Positive → one unnecessary inspection. Annoying, but cheap.

False Negative → production downtime, secondary damage, possible injury

Specific Metrics

F_1- Score

The Harmonic mean.
Composed of Precision and Recall

- Sensitivity (Recall) – Measures the actual failures. “Of all the anomalies, how many did the model detect?”
- Efficiency (Precision) – Measures the Total Alarms raised by the model.

ROC-AUC

Area under the Receiving Operating Characteristic Curve.

This metric measures the model’s ranking ability.

“If pick one broken pump and one normal pump, what is the probability that the model give the broken one a higher risk?”

.

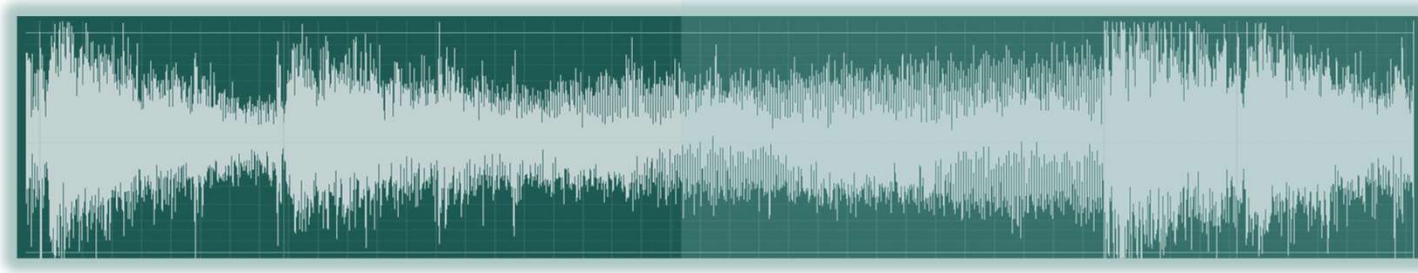
Building by Putting out Fires

Every feature of the final architecture exists because an earlier versions crashed and burned

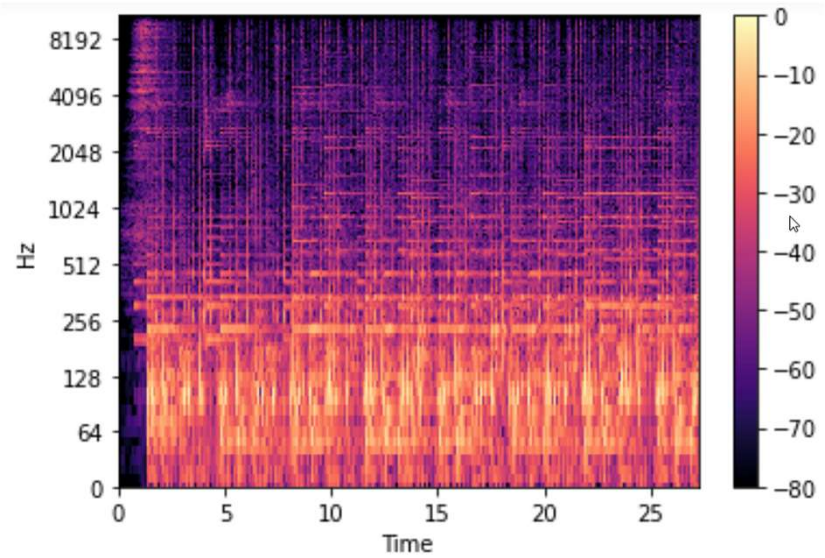
Raw Audio is too Volatile

A **One Dimension Waveforms** are incompatible with a CNN model.

- High Dimensional
- Noisy
- Lacks spatial structure



The model is incapable of finding the pattern.



Convert each image to a **Two- Dimensional Log-Mel Spectrogram** *(128 x 313)*

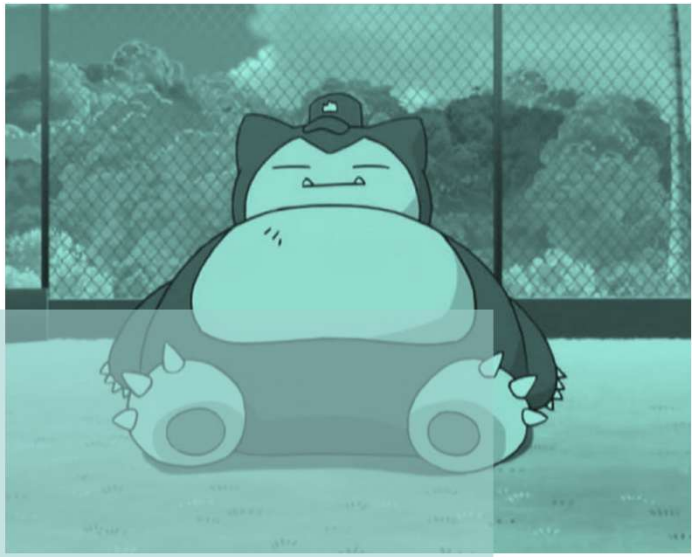
Trade – offs:

- **Window (N_FFT = 2048)** – the focus lens for each frame.
- **Stride (hop = 512)** – 75% overlap, like a flipbook.
- **Mel scale (128 bins)** – spends pixels on the low frequencies where mechanical faults live.
- **Decibel (log) scaling** – keeps loud bursts from drowning out subtle patterns.

Out-of-Memory Crash (OOM)

Loading all 4.200+ Spectrograms into RAM at once.. **Kernal Kill.**

- The dataset is small as audio, but each spectrogram is 160kb



Lazy Loading

Holding only the file paths in memory. **Post codes, not photos**.
Batch load from drive on demand and discard it after use.

Trade offs:

- Reading from the drive is significantly slower than RAM
- Same code on 24, 4000, 4,000,000 million files.
- Per-sample standardization.

Degenerative Model

Factory Useless:
With only ~11% of the audio clips classified as abnormal, how can we stop the model from guessing normal all the time and getting an 89% accuracy with 0% Recall?

The Industry chosen failed Fix: **SMOTE**
Increase the number of abnormal clips by producing synthetic audio chimeras. What does the model learn?

Inverted-Frequency Weighted Loss

Tax ~~the RICH~~ the model asymmetrically: Missing an anomaly "hurts" approximately **8.2x more than a false positive**

Replicates real factory cost asymmetry, the model is forced to take rare classes seriously.

$$w_c = \frac{N_{total}}{k \cdot N_c}$$

w_c = The final "fine

N_{total} = The total number of audio clips in your entire training dataset

k = The total number of categories you are trying to predict

N_c =The number of audio clips that belong specifically to class

The Micromanager

Overfitting

The original model ended in a Flatten layer → 2.5 million parameters.

- The linear head in CNN needs a 1D vector.

For 319 broken pump training examples, however, that is not good. **The model memorised the training, instead of Generalising and learning the pattern.**

Ruins Out-Of-Sample use case

Global Average Pooling

Replace the Flatten Layer with a GAP: we collapse each feature map to a single average.

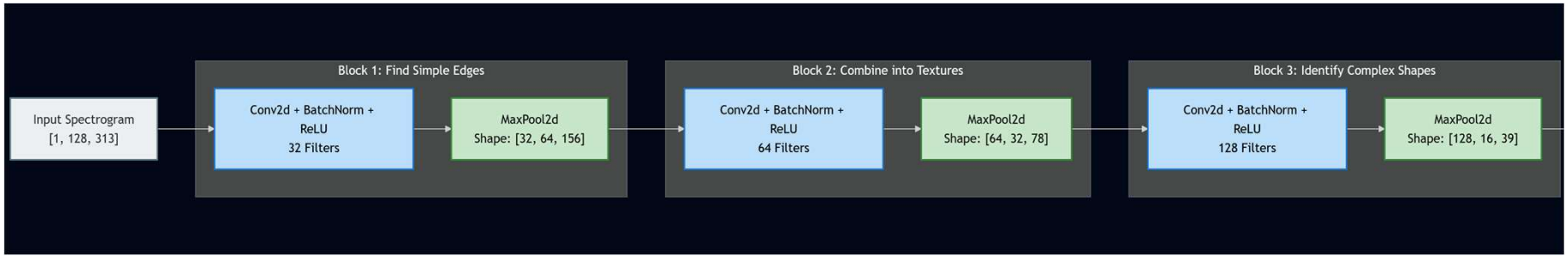
- A specific frequency band squished down to a single average number.

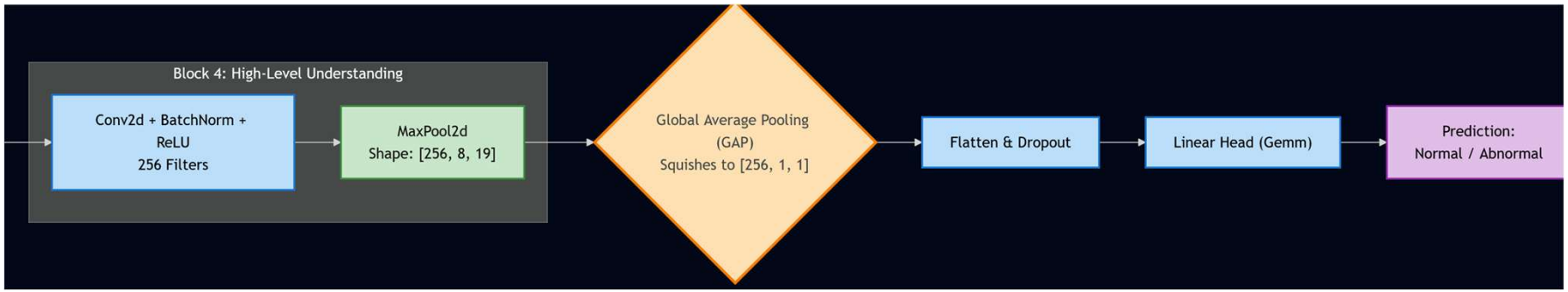
"Did you hear a grinding noise anywhere in the 10 second clip?"

Destroys the information on where the sound happened, the model is forced to learn broad generalised rules.

Parameters: ~2.5M → ~390k at BF=32 (a **95% reduction**).

The Model

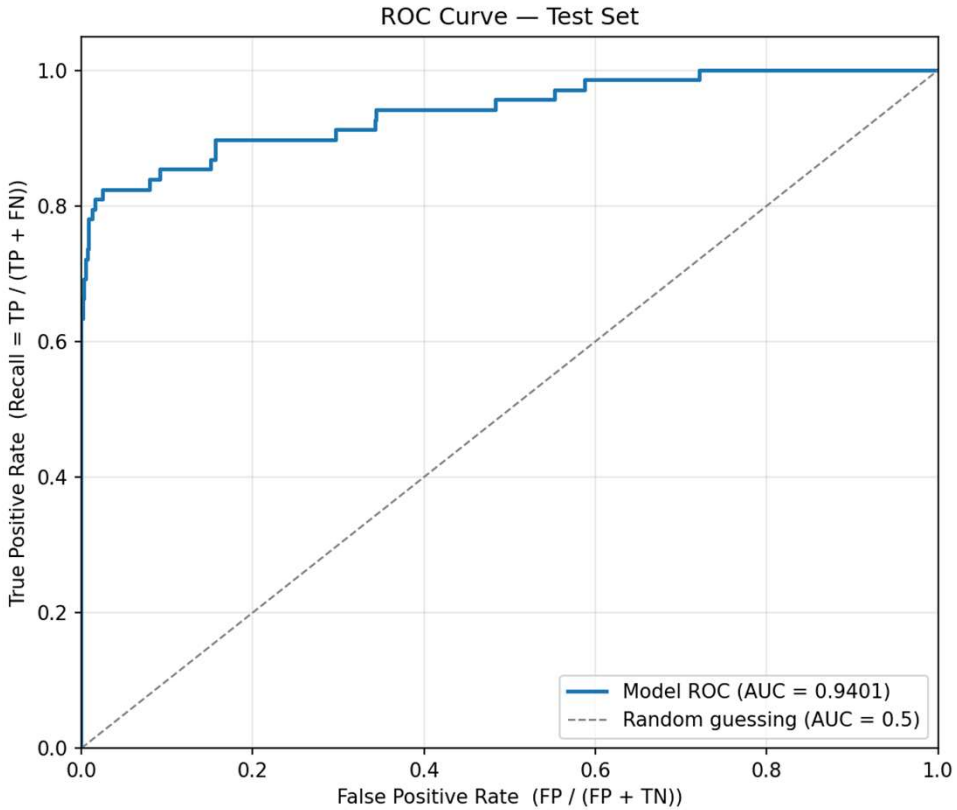

Input Spectrogram
[1, 128, 313]
Block 1: Find Simple Edges
Conv2d + BatchNorm + ReLU
32 Filters
MaxPool2d
Shape: [32, 64, 156]
Block 2: Combine into Textures
Conv2d + BatchNorm + ReLU
64 Filters
MaxPool2d
Shape: [64, 32, 78]
Block 3: Identify Complex Shapes
Conv2d + BatchNorm + ReLU
128 Filters
MaxPool2d
Shape: [128, 16, 39]

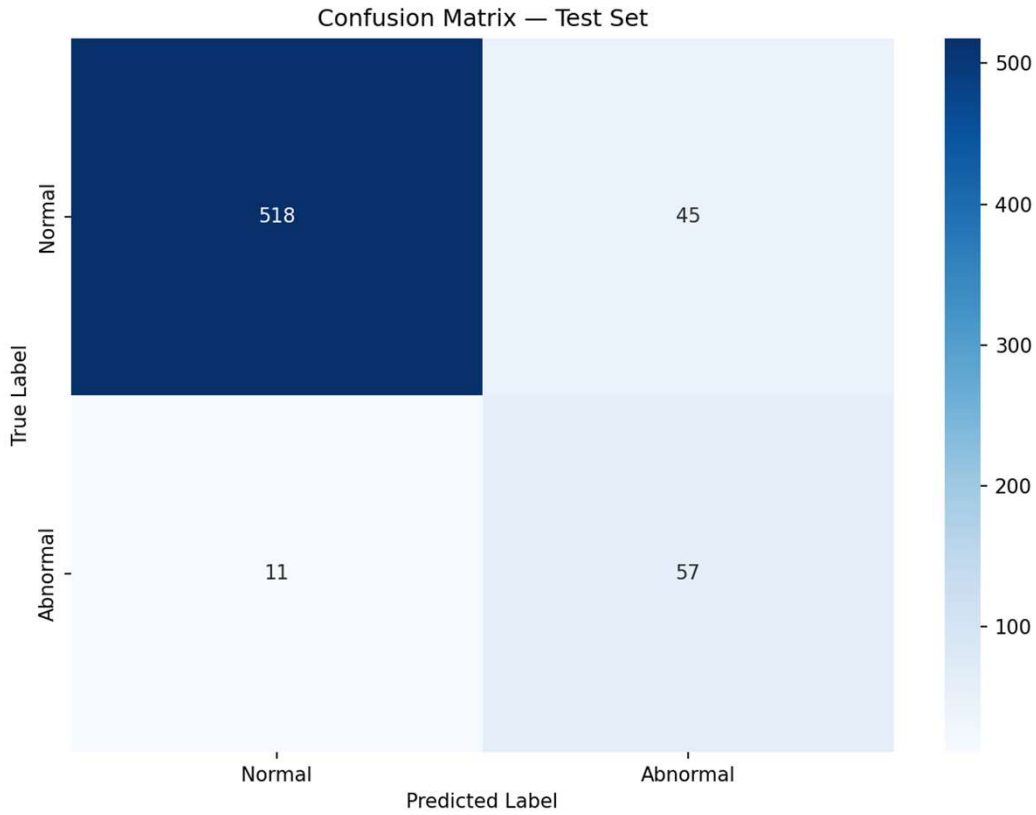

Block 4: High-Level Understanding
Conv2d + BatchNorm + ReLU
256 Filters
MaxPool2d
Shape: [256, 8, 19]
Global Average Pooling (GAP)
Squishes to [256, 1, 1]
Flatten & Dropout
Linear Head (Gemm)
Prediction:
Normal / Abnormal

Did it Work?

04

Yes


ROC Curve — Test Set
True Positive Rate (Recall = TP / (TP + FN))
False Positive Rate (FP / (FP + TN))
0.0
0.2
0.4
0.6
0.8
1.0
Model ROC (AUC = 0.9401)
Random guessing (AUC = 0.5)


Confusion Matrix — Test Set
True Label
Predicted Label
Normal
Abnormal
518
45
11
57
500
400
300
200
100

Hyper-Parameter Optimisation

A 12-point Cartesian Grid Search:
Learning rate x Batch size X base filter

Large batch + Aggressive learning rate sent F1 score to collapse and increased false positives by 6x

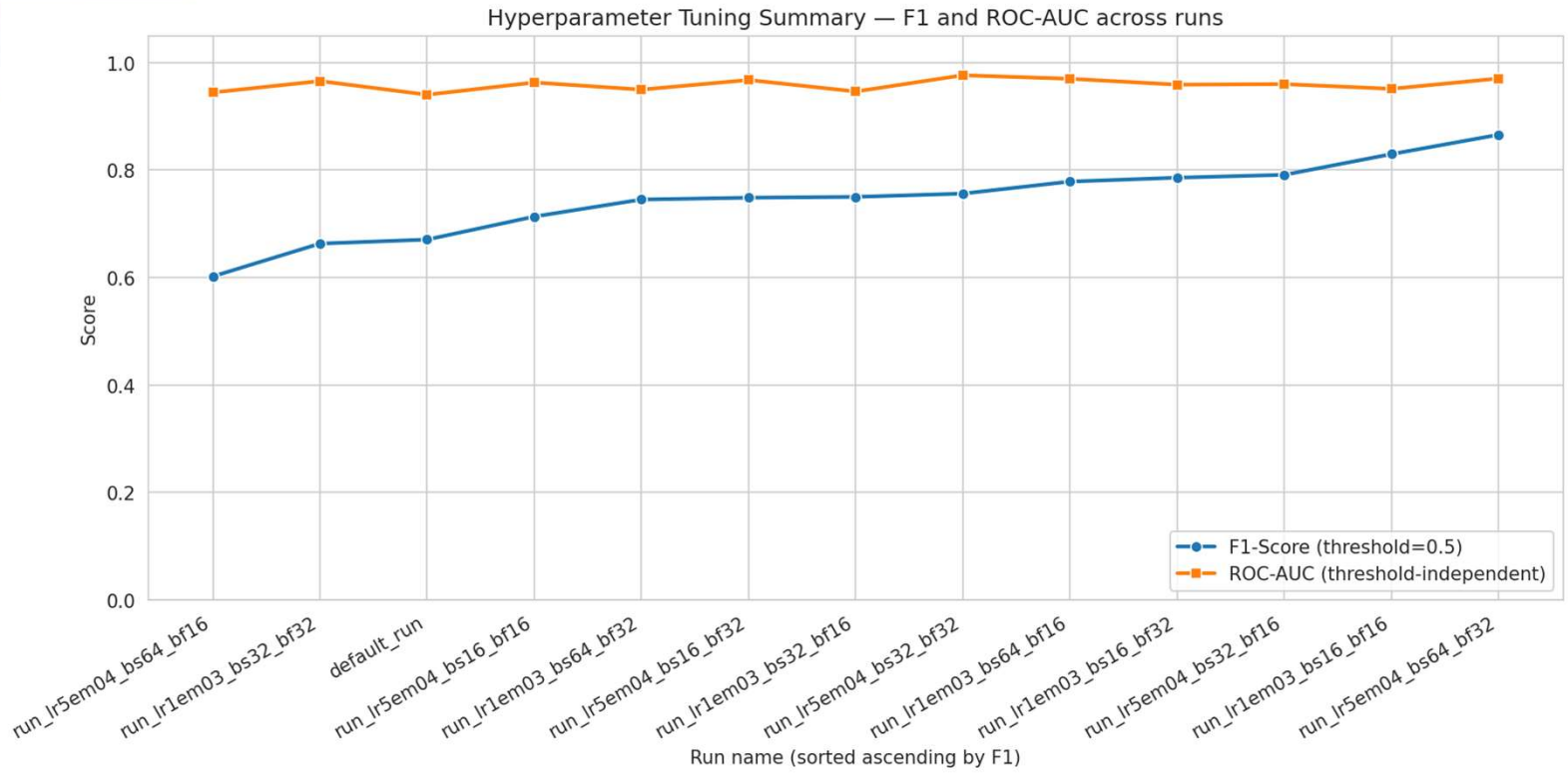
The Mechanism

Big batch (64) → a smoothed, low-noise gradient – a confident view of the descent direction.

Aggressive LR (1e-3) → big steps in that direction.

Together: a big step in a confidently wrong direction – the optimizer overshoots into a wide, shallow basin with a fuzzy decision boundary

The Champion Configuration



Learning rate	0.0005
Batch Size	64
Base Filter	32

Headline Results

Test set

631 recordings

F1-Score

0.866
Not a 1, but close

ROC-ACU

0.9703

Recall

0.853
Of every 100 failures the model catches 85%

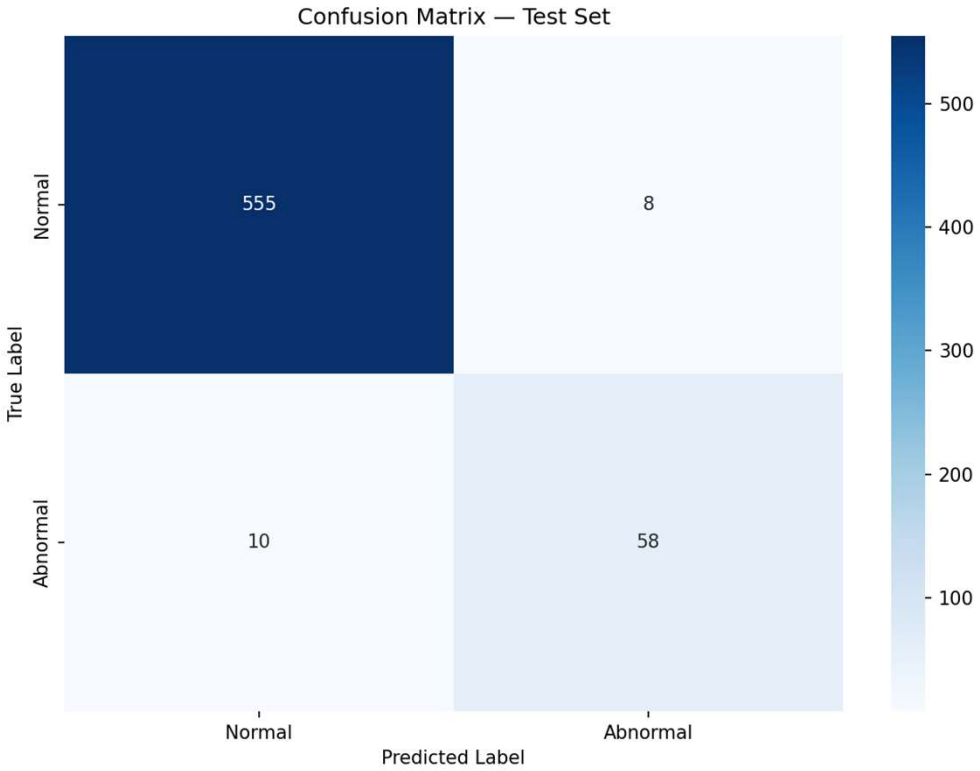
Precision

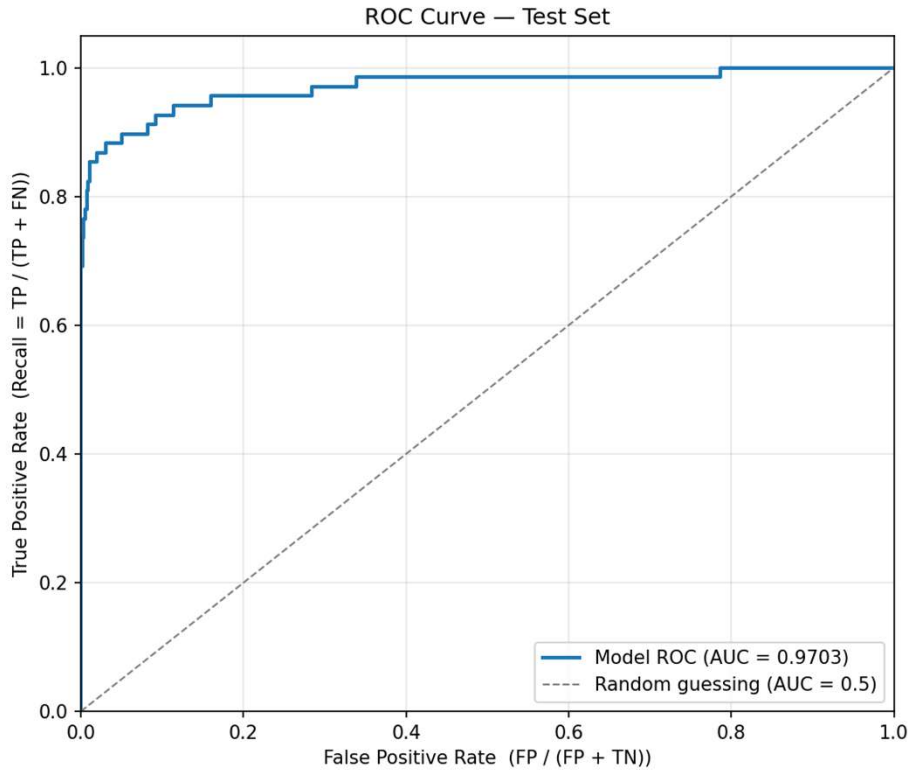
0.879
~12% of alarms were false positives

Upshot

The model catches: 58 of 68 truly broken pumps and raises only 8 false alarms across 563 normal recordings

Confusion Matrix & ROC Curve



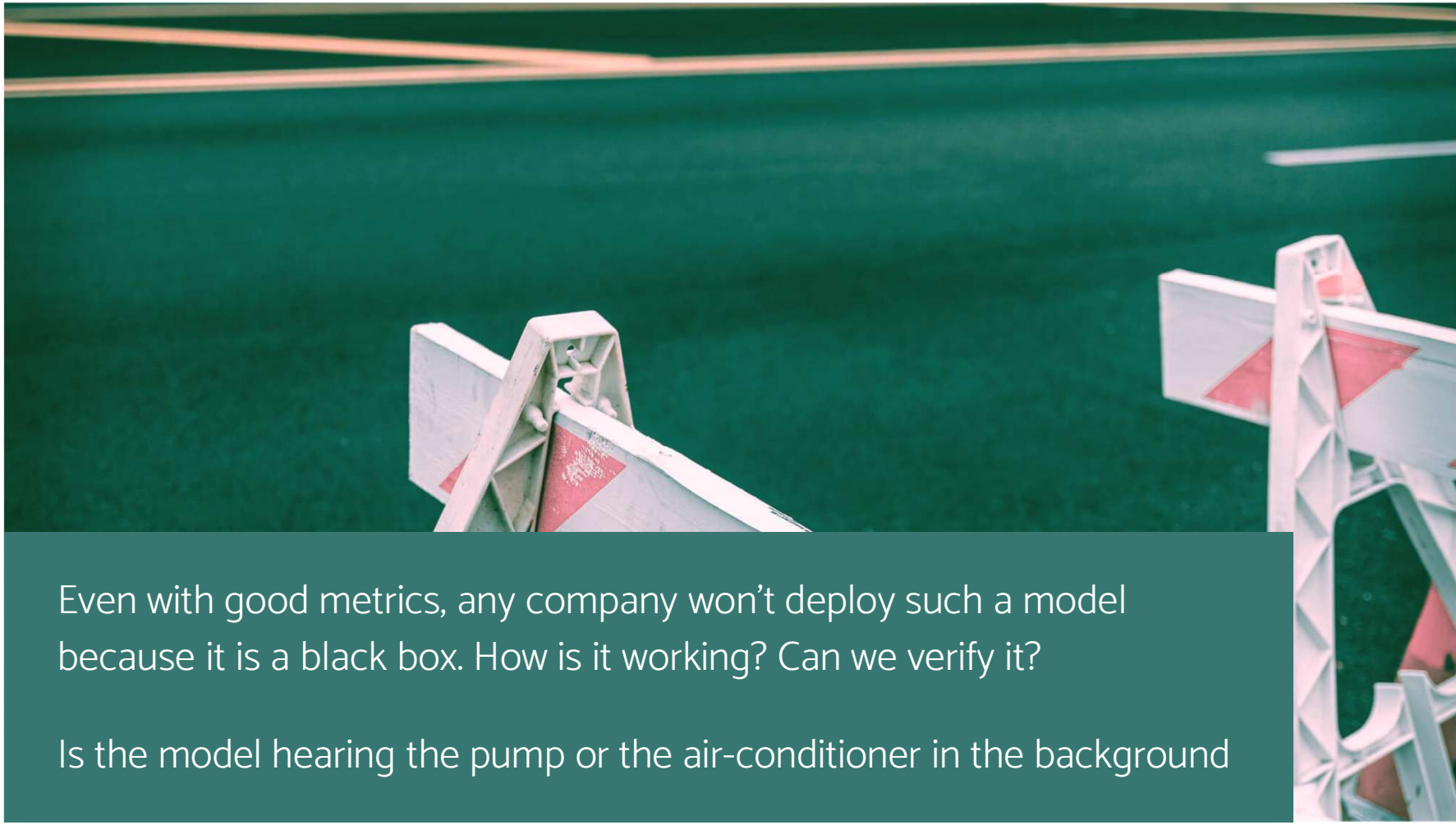


Healthy correctly cleared: **555 / 563**
Broken correctly caught: **58 / 68**
False-Positive rate: **1.4%** · False Negative: 10

Near-vertical climb on the left reaches ~85% catch rate at essentially zero false alarms.

Plateaus around 97% catch rate as the threshold relaxes.

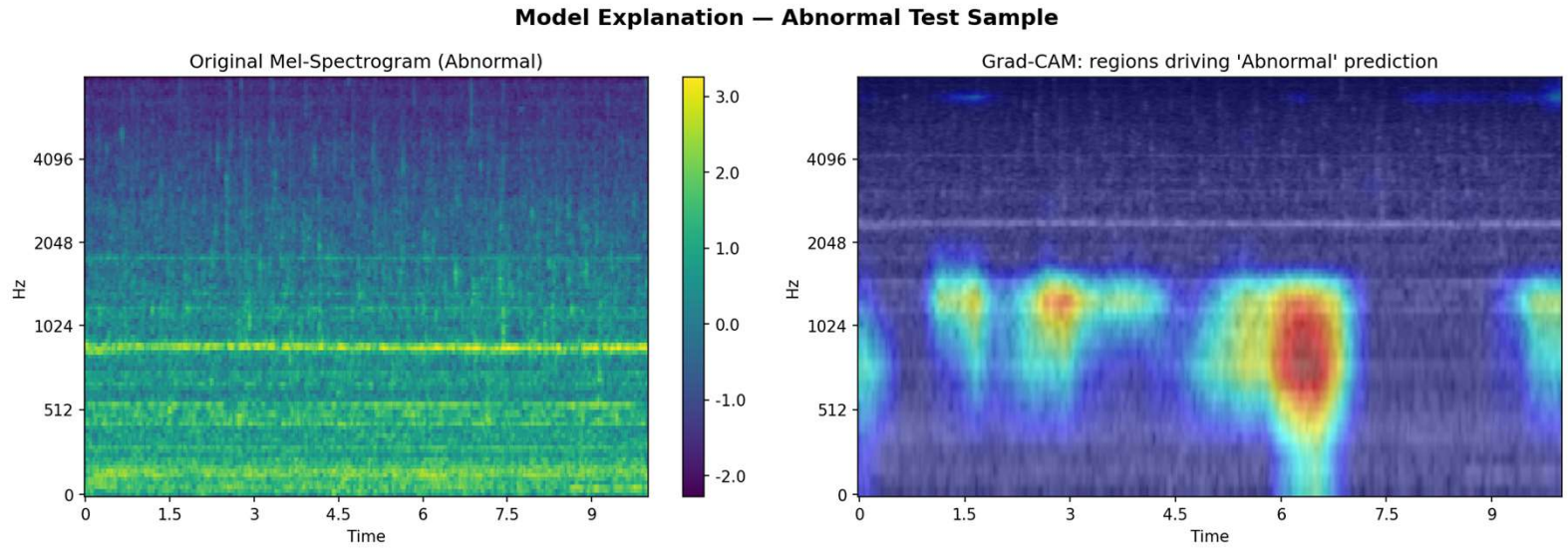
The Black Box



Even with good metrics, any company won't deploy such a model because it is a black box. How is it working? Can we verify it?

Is the model hearing the pump or the air-conditioner in the background

GRAD- Cam



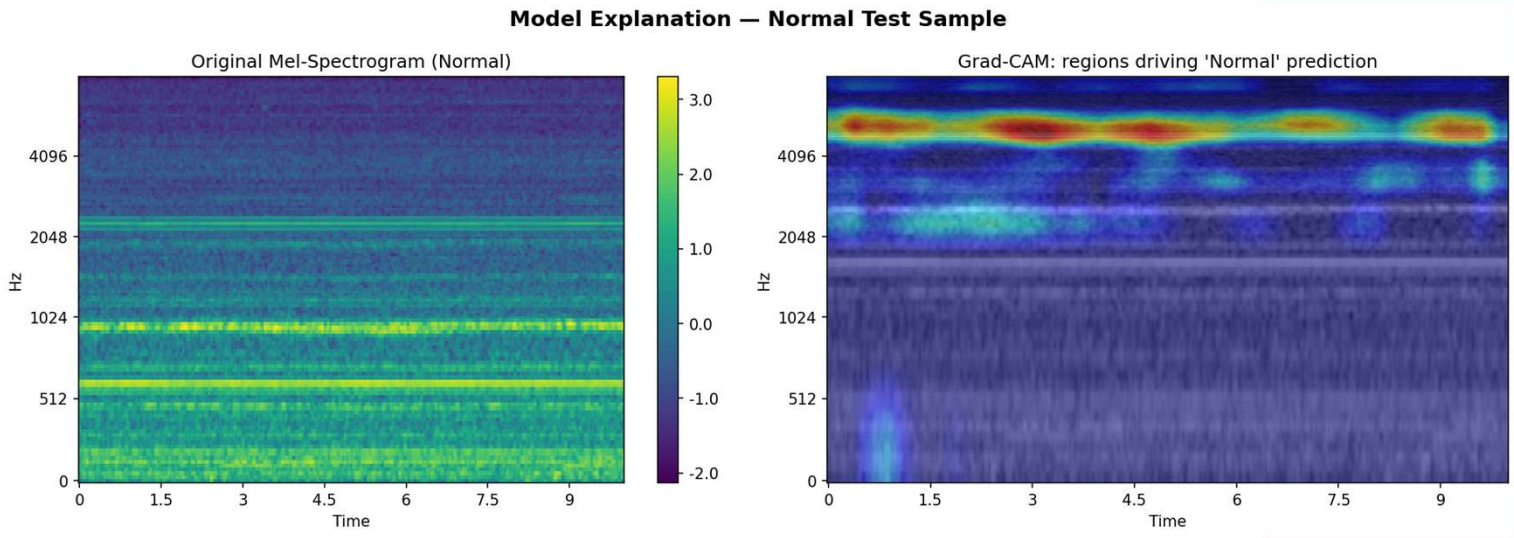
Work backwards through the model to highlight which spectrogram regions drove each decision.

Heatmap "X-ray" of the model's attention.

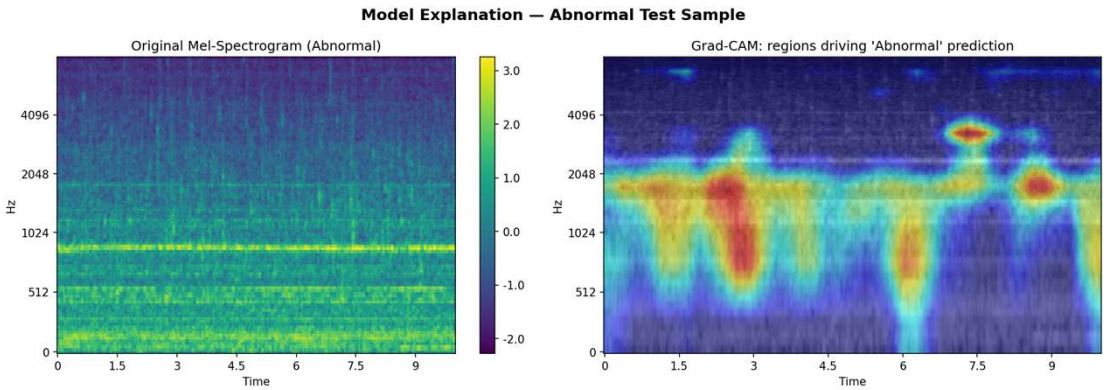
Abnormal: localized mid-frequency bursts (~512–2,048 Hz) at ≈1.2 s, 4.0 s, 6.0 s – bearing wear & cavitation signatures.

Normal: broad time-uniform band above ~4 kHz – the stationary noise floor of an undamaged pump.

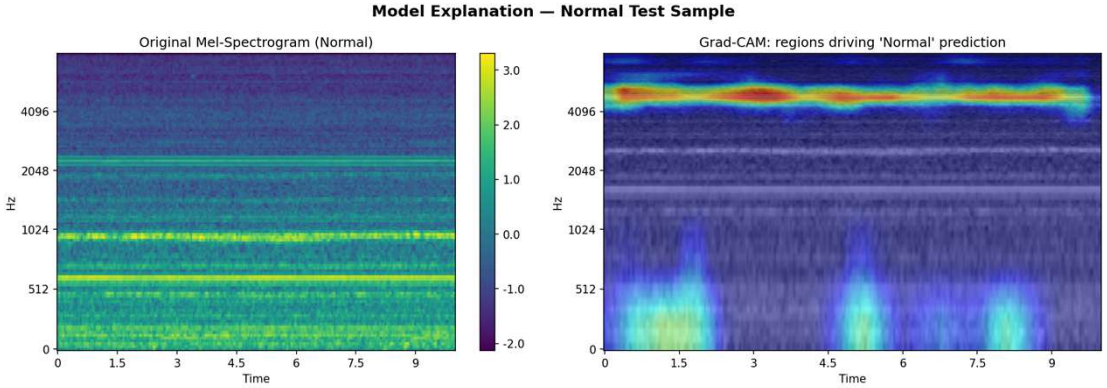
.

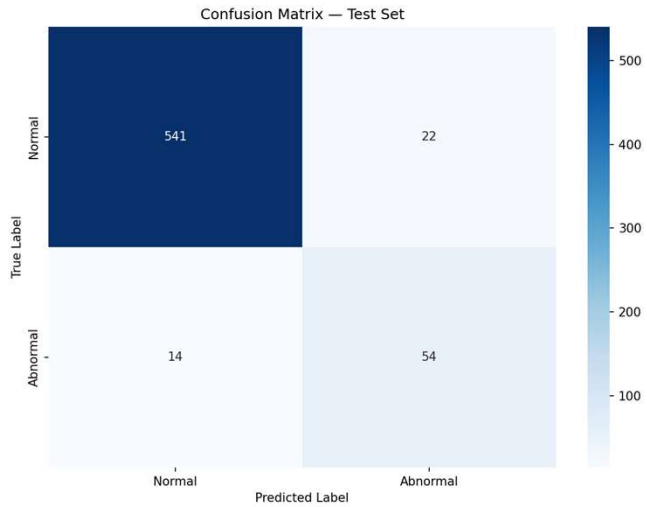


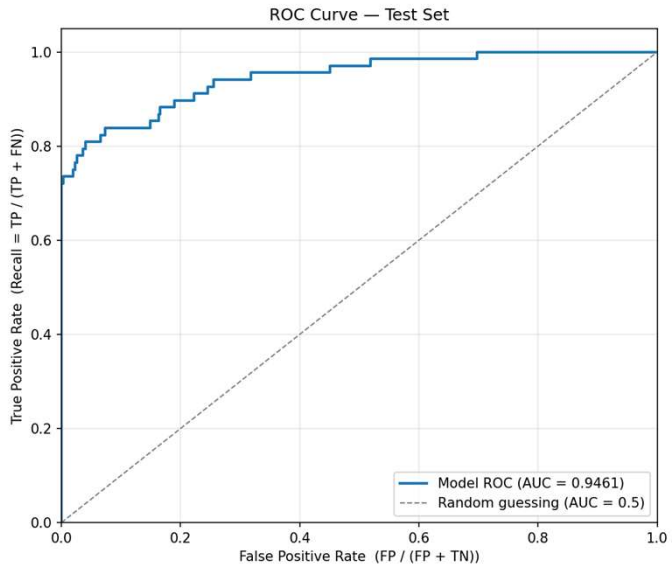
GRAD- Cam



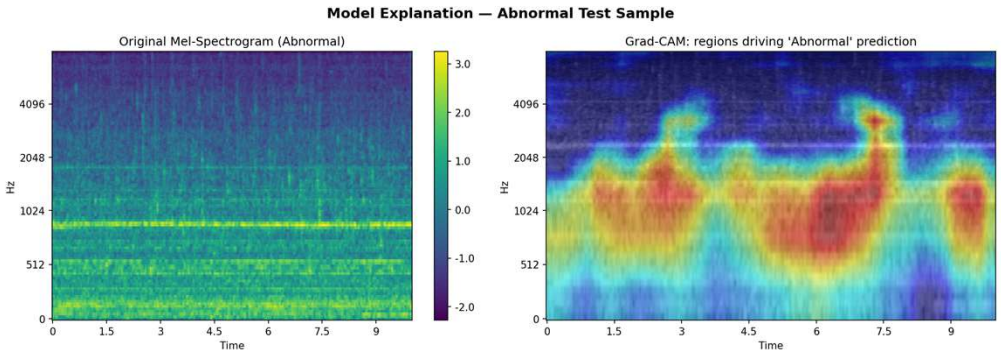
Learning rate	0.003
Batch Size	32
Base Filter	16



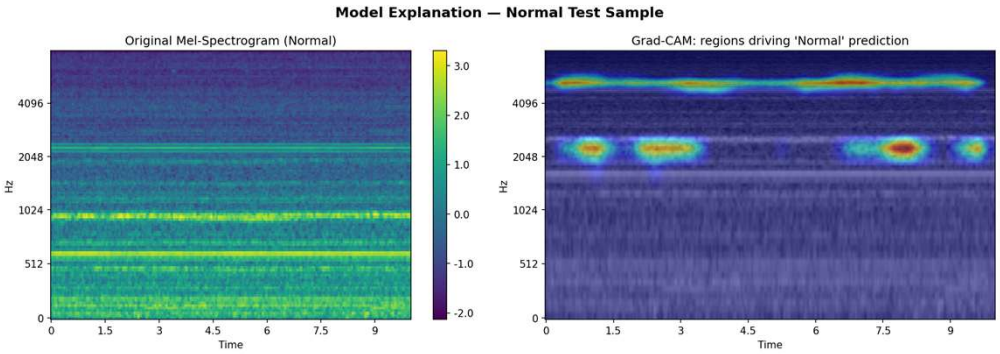


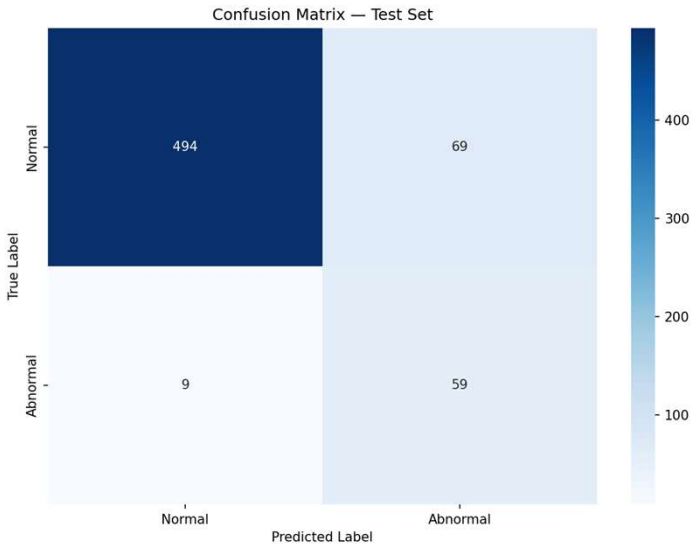


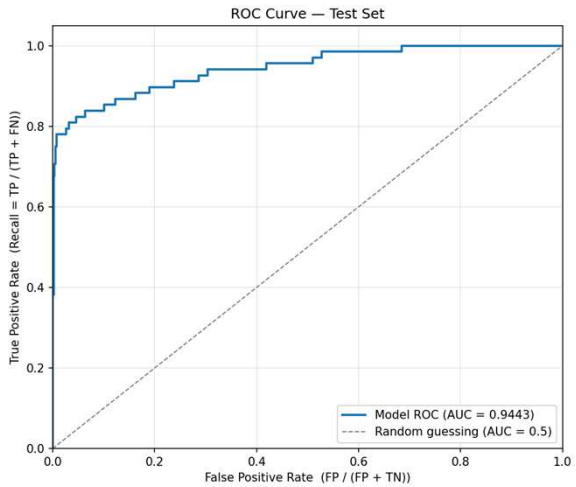
GRAD- Cam



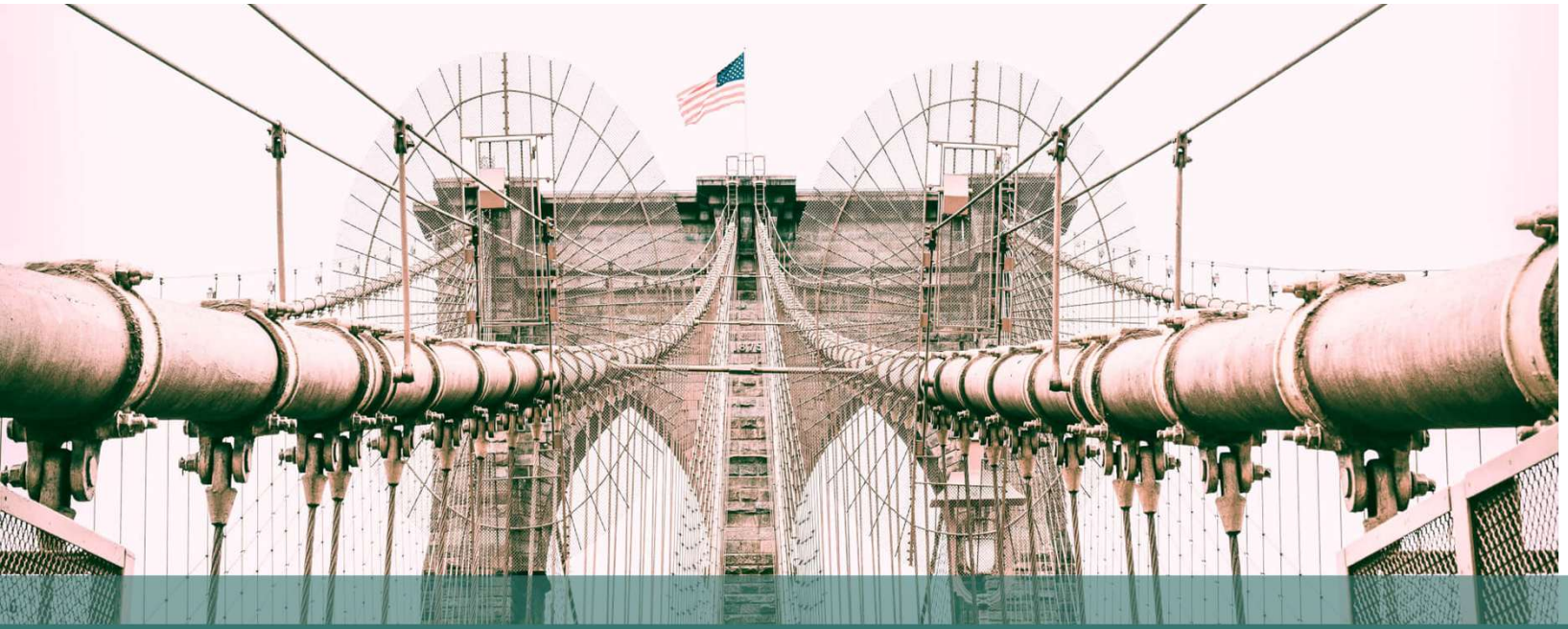
Learning rate	0.0005
Batch Size	64
Base Filter	16







Is It Over Yet?



Limitations

Domain Shift: Model is trained on a specific noise profile.

Single Sensor Reliance: One signal to learn. Real systems fuse audio with vibrations, thermal, and current data

ROC Threshold: A business decision

Future Work

K-fold cross-validation: Turn pointt estimates into confidence intervals

Spec-Augment: Data augmentation to go beyond that 85% recall ceiling

Conclusion

1. Weighted loss over synthetic data.
2. Global average pooling
3. 12-point Cartesian grid
4. Keep pushing the grid to find better performance (BS=64) What would 128 results be?

Self-Evaluation

What went well

Major features and biases I planned to implement and address were solved

Treated each failure as a hypothesis test, and every architecture choice has a reason behind it.

Prevented early data leakage

What I would do differently?

Run a K-fold from the start.

Experiment with Audio Spectrogram Transformer or PANNs

Build a test suit from day one rather than retro-fitting it.

What didn't?

Learning about audio, spectrograms, and how to convert it

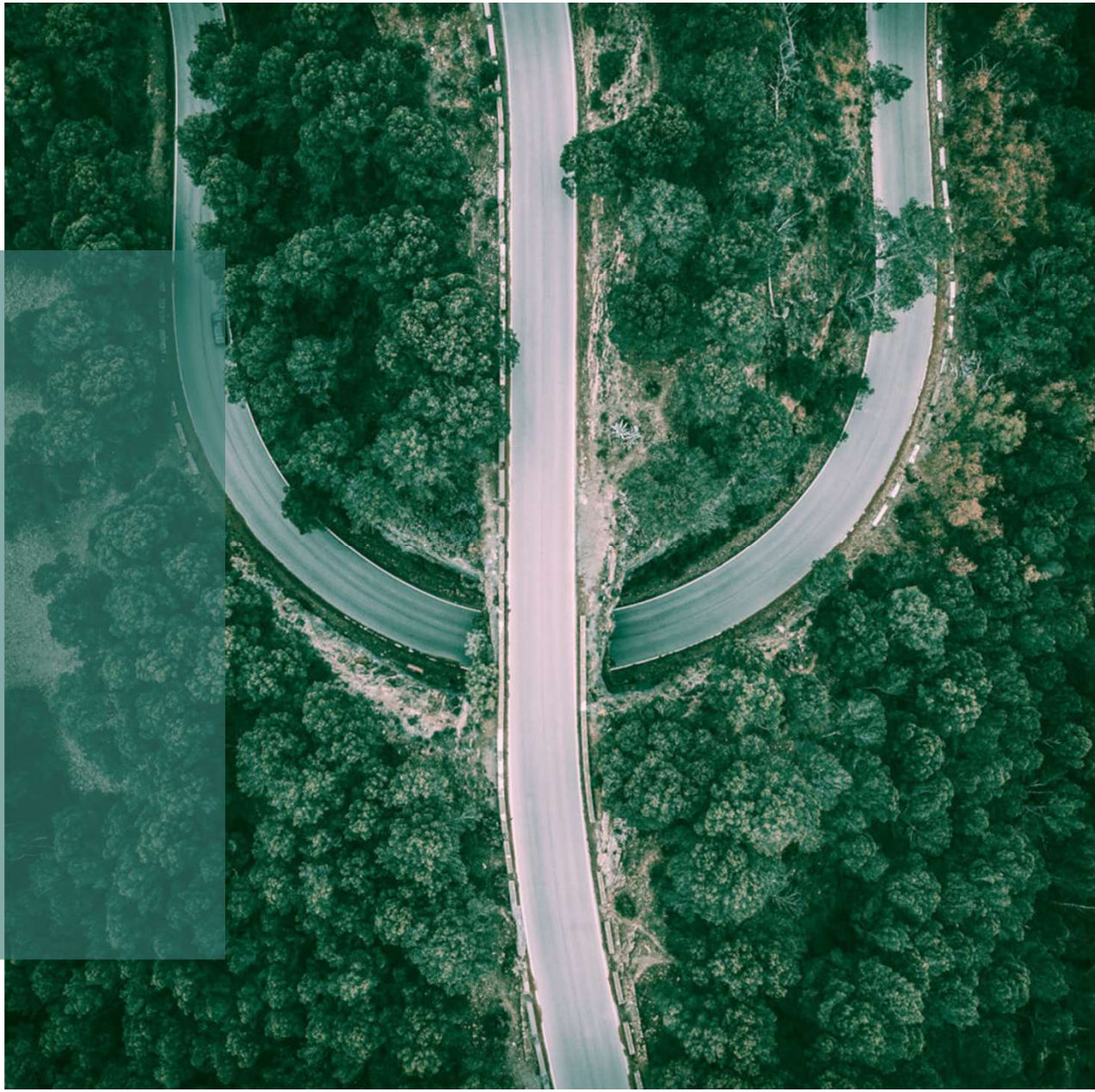
The calculus of GRAD-Cam

What went wrong

Nope, Not Yet.

THANKS

I will not be accepting any questions at this time.



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